PROJECT 2 - STATISTICAL LEARNING

Daniela Bialy 100550983 - Elena Lardies 100550964

1.Data preprocessing & visualization tools

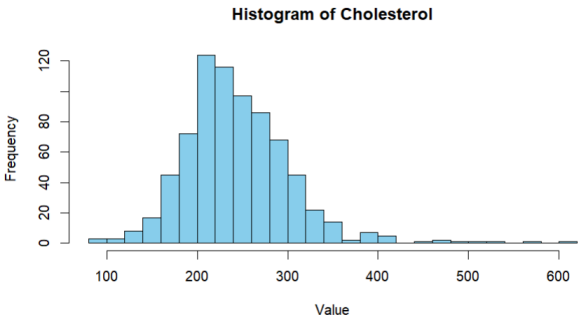
This project focuses on predicting the presence of Heart Disease (a categorical target variable). The data used for it is called “heart.csv” which initially contains 12 variables and 918 observations. The variables are: **Age** (patient's age in years), **Sex** (M/F), **ChestPainType** (ATA/NAP/ASY/SYM categories), **RestingBP** (resting blood pressure in mm Hg), **Cholesterol** (serum level in mg/dl), **FastingBS** (binary >120 mg/dl), **RestingECG** (Normal/ST/LVH), **MaxHR** (maximum heart rate during stress test), **ExerciseAngina** (Y/N), **Oldpeak** (ST depression level), **STSlope** (Up/Flat/Down), and **HeartDisease** (binary outcome: 1=disease, 0=no disease).

Initial data exploration of heart.csv revealed only 4 missing values (NAs), which were promptly removed. However, 172 zero values in Cholesterol were identified as implausible (medically impossible), so they were converted to NAs. Skewness analysis confirmed right-skewness, leading to KNN imputation for realistic value estimation. We also factorize categorical variables.

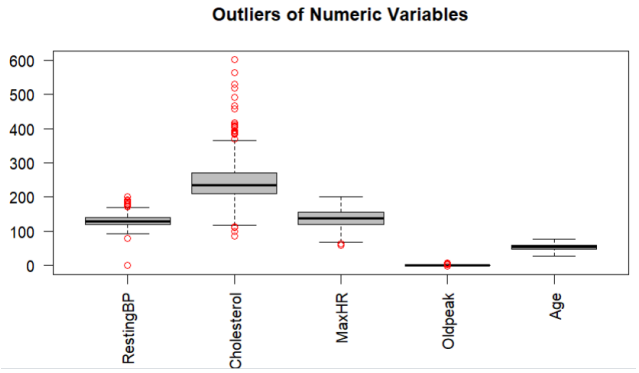
```
> table(is.na(data_original))

FALSE   TRUE
11012      4

> sum(data$Cholesterol == 0)
[1] 172
```

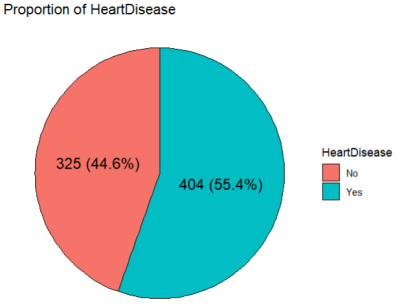


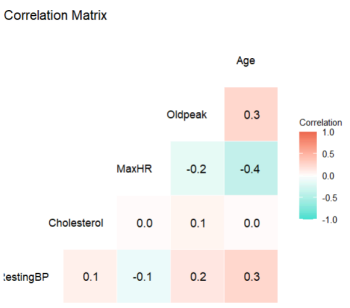
Analysis continued by examining outliers in numeric variables. Rather than applying a method to deal with outliers, we preferred to leave them as is, excluding only clearly impossible or clinically inconsistent values, which resulted in just 3 rows removed as erroneous data. Many points far from the median remain physiologically plausible and were retained, since extreme realistic values like very high blood pressure or cholesterol provide valuable predictive information for heart disease models.



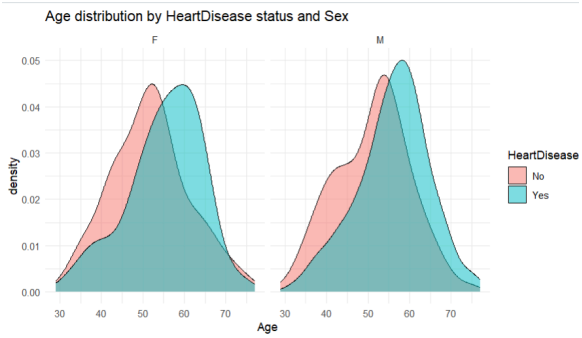

```
print(outliers)
RestingBP Cholesterol       MaxHR     Oldpeak         Age
       28          28           2          16           0
```

We then split the dataset into training (80%) and test sets, and generated exploratory plots to better understand the data patterns (additional plots and explanations available in the HTML.

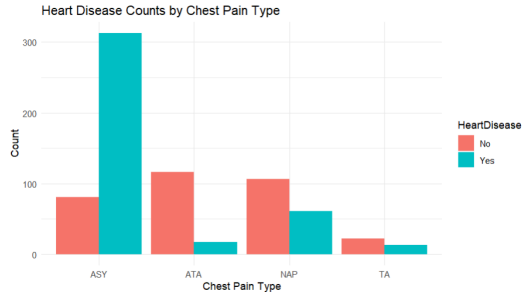




A pie chart of the target variable shows that 55.4% of patients have heart disease while 44.6% do not. The correlation matrix was then computed for the numeric variables, showing that Age is moderately related to MaxHR, Oldpeak, and RestingBP, while the remaining features exhibit very weak linear relationships, so numeric variables are mostly independent from each other.

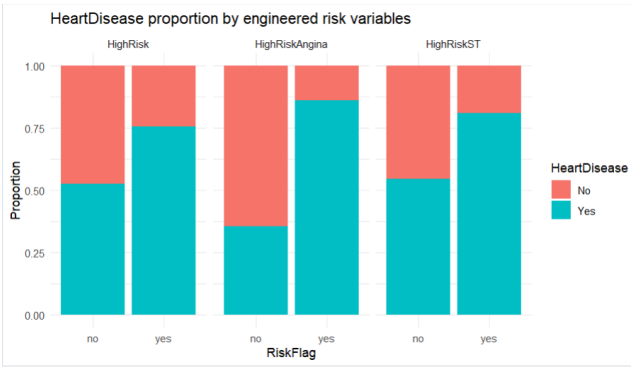


The faceted age-density plot shows that, for both women and men, patients with heart disease tend to be older on average than those without, even though their age distributions overlap substantially. In addition, there are clearly more men than women in the HeartDisease = “Yes” group, indicating that heart disease is more frequent among male patients in this dataset.



The plot shows that the probability of having heart disease varies strongly by chest pain type. Patients with ASY chest pain are mostly in the HeartDisease = “Yes” group, while those with ATA are predominantly “No”, and NAP/TA lie in between, indicating that chest pain type is a very informative predictor of heart disease status.

New binary variables were created to better capture high-risk profiles in the data. **HighRisk** takes value “yes” when a patient has RestingBP greater than 140 and Cholesterol above 240. **HighRiskAngina** is “yes” for patients whose ChestPainType is ASY or NAP and who also present ExerciseAngina = “Y”. **HighRiskST** is “yes” when ST_Slope is “Down” and Oldpeak exceeds 2, indicating marked ST-segment depression during exercise.

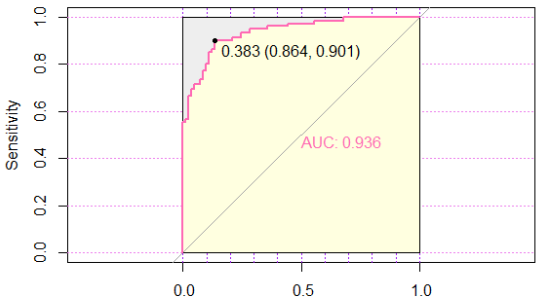


Across all three engineered risk flags, the “yes” category shows a much higher proportion of patients with heart disease than the “no” category. This means that when any of these risk flags is present, the probability of having heart disease increases substantially compared with patients without that risk flag, indicating that the new variables are meaningful and useful predictors for HeartDisease in the model.

2.Classification: emphasis on prediction

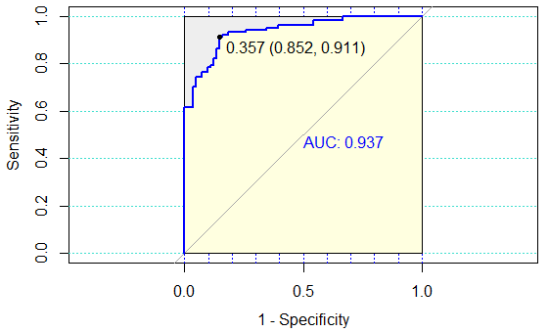
Logistic regression

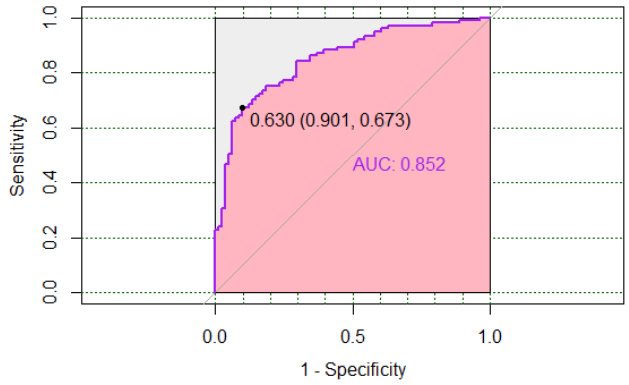
The standard logistic regression model shows excellent predictive performance, with **AUC ≈ 0.94**.The ROC curve lies close to the top-left corner, reflecting high sensitivity and specificity across a wide range of thresholds. Using the optimal cut-off, the model achieves a good balance between correctly identifying diseased patients and minimizing false positives.



Penalized logistic regression

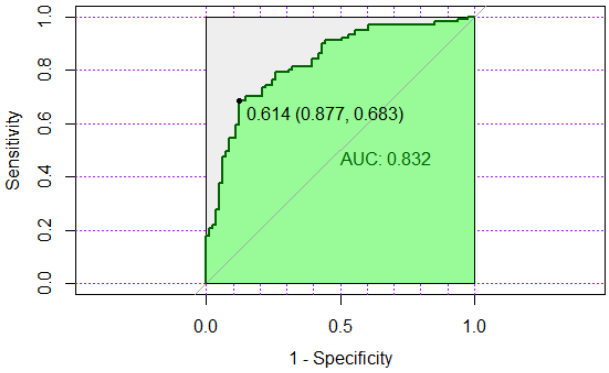
Penalized logistic regression achieves almost the same performance, with an **AUC** also around **0.94**. By applying regularization, the model shrinks less informative coefficients toward zero and highlights the most important predictors, improving stability and reducing potential overfitting.





LDA (Linear Discriminant Analysis)

LDA uses only continuous predictors (Age, RestingBP, Cholesterol, MaxHR, Oldpeak). It yields an **AUC ≈ 0.85**, indicating good but clearly inferior performance compared with the full logistic regression. The ROC curve shows that at an optimal threshold around 0.63, LDA attains high sensitivity (0.90) but more moderate specificity (0.67).

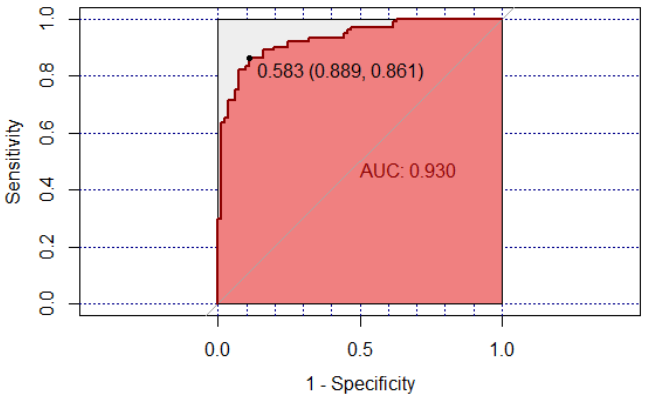


QDA (Quadratic Discriminant Analysis)

QDA assumes that the data within each class follows a normal distribution, but unlike LDA, it allows each class to have its own covariance matrix. Trained on the continuous predictors, produced an **AUC ≈ 0.83**, showing good but lower discriminative power than logistic regression and LDA. The optimal threshold (0.61) yields high sensitivity (0.88) but moderate specificity (0.68). QDA's flexibility helps capture nonlinear patterns, though it may overfit when sample sizes are limited.

Naive Bayes

Assumes independence between variables given the class and uses Bayes' theorem to compute probabilities.The model achieves **AUC ≈ 0.93**, indicating excellent predictive ability. At the optimal probability threshold (0.58), it correctly identifies 88.9% of patients with heart disease and 86.1% of those without.Despite its simplicity and strong independence assumption, Naive Bayes performs nearly as well as logistic regression — showing it captures key patterns in the predictors effectively.



KNN

KNN classifies each observation based on the majority class of its k nearest neighbors in the feature space. The manually tuned KNN achieved an accuracy of **0.71978** on the test set, with k = 4 selected as the best value. Using 10-fold cross-validation, the KNN model selected k = 9 and achieved a slightly lower accuracy of **0.68743**. The manually tuned KNN achieved a slightly higher accuracy on the test set; however, the cross-validated KNN provides a more reliable estimate of performance, as it averages results across multiple training and validation splits. Overall, KNN shows moderate predictive performance and performs worse than logistic regression models considered earlier.

Conclusions

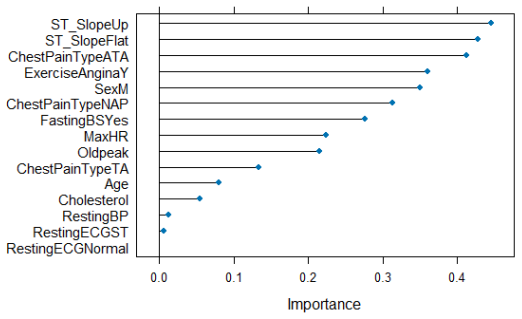
Across all models, predictive performance was strong but varied in accuracy and stability. Logistic regression achieved the best overall results, with an accuracy of 0.87 and a high Kappa of 0.74, consistent with its excellent discrimination (AUC ≈ 0.94). Penalized logistic regression performed similarly, confirming that reducing overfitting through regularization improves robustness without sacrificing predictive power. Naive Bayes also showed strong performance (accuracy ≈ 0.86, Kappa ≈ 0.72, AUC ≈ 0.93) despite its simplifying feature independence assumption. LDA and QDA

achieved lower accuracy (0.75–0.76) and moderate Kappa values (0.49–0.51), reflecting weaker discrimination. KNN showed only moderate performance, with test accuracy around 0.72 when manually tuned and about 0.69 under cross-validation, performing clearly worse than the Logistic and Naive Bayes models.

3.Classification: emphasis on interpretation

It is appropriate to focus the interpretation mainly on the standard and penalized logistic regression models, as they are the best-performing methods and also the most interpretable, along with Decision Trees and Random Forest, which will be discussed below.

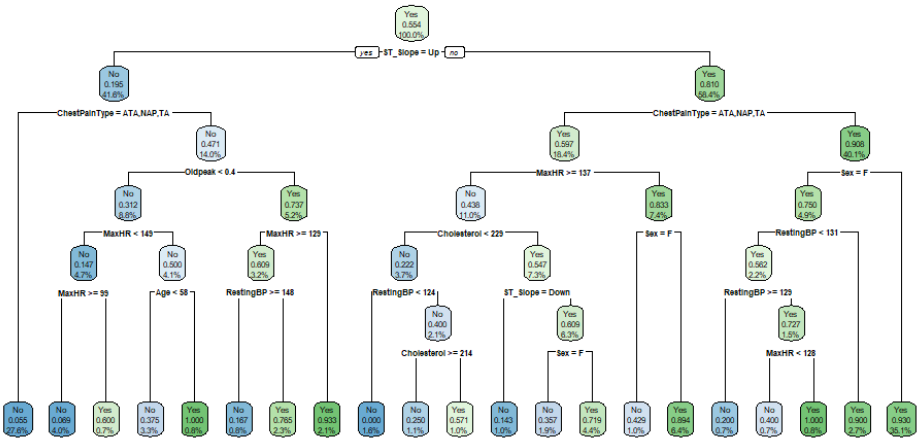
```
Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)     -2.431346   1.674313  -1.452 0.146461
Age              0.014951   0.014793   1.011 0.312168
SexM             1.590174   0.314418   5.058 4.25e-07 ***
ChestPainTypeATA -1.992311   0.371558  -5.362 8.23e-08 ***
ChestPainTypeNAP -1.536730   0.284149  -5.408 6.37e-08 ***
ChestPainTypeTA  -1.655965   0.488211  -3.392 0.000694 ***
RestingBP        0.003943   0.007036   0.560 0.575240
Cholesterol      0.003753   0.002312   1.623 0.104582
FastingBSYes     1.213729   0.298501   4.066 4.78e-05 ***
RestingECGNormal 0.073155   0.297117   0.246 0.805515
RestingECGST     0.156299   0.380765   0.410 0.681449
MaxHR           -0.009210   0.005387  -1.710 0.087348 .
ExerciseAnginaY  0.814697   0.265485   3.069 0.002150 **
Oldpeak          0.303286   0.130821   2.318 0.020431 *
ST_SlopeFlat     1.466068   0.459159   3.193 0.001408 **
ST_SlopeUp      -0.946786   0.482053  -1.964 0.049522 *
```

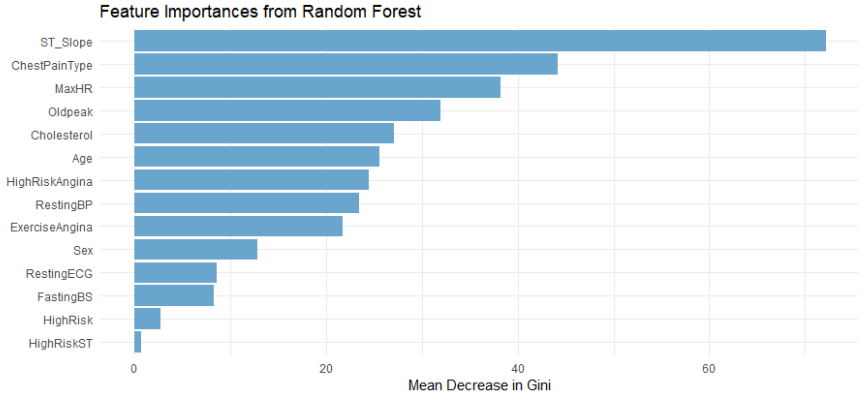


If we check the logistic regression summary, the Estimate column shows the direction and strength of each predictor's effect: positive estimates increase the probability of heart disease, while negative ones decrease it. The p-values (Pr(>|z|)) and significance stars indicate which variables are statistically important. In this case, variables such as ***ST slope, ChestPainType, ExerciseAngina, Sex, FastingBS***, and ***Oldpeak*** are significant and show strong effects. This is consistent with the Penalized Logistic Regression importance plot, where the same variables appear as the most influential. Less important predictors (e.g., ***Cholesterol*** or ***RestingBP***) have weak significance and low importance in both models. Together, these results confirm that logistic regression—especially with regularization—provides the best balance between predictive accuracy, robustness, and interpretability.

Decision Tree

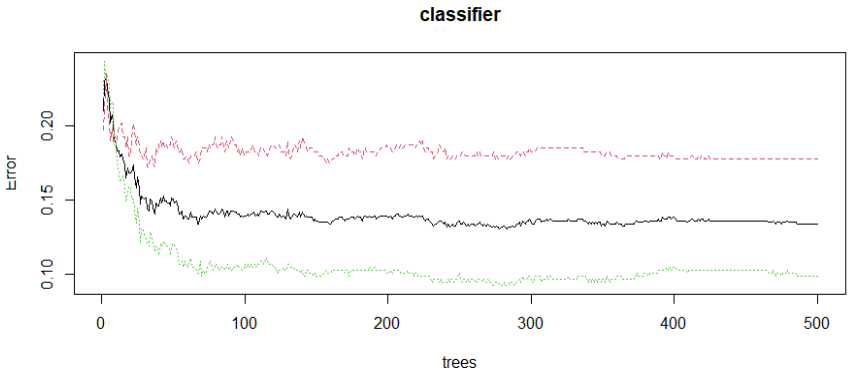
The decision tree offers an interpretable, rule-based representation of heart disease risk that complements the earlier parametric models. The first split on ***ST_Slope*** confirms its importance, consistent with the logistic and penalized logistic regression results. Subsequent splits on ***ChestPainType, Oldpeak, MaxHR***, and ***Age*** reveal nonlinear interactions between clinical variables that are not captured by simpler models. The tuned tree achieves an accuracy of about **0.85**, which is slightly lower than Logistic Regression and Naive Bayes but still competitive. Overall, the decision tree trades some predictive accuracy for transparency, making it useful for understanding decision rules and clinical thresholds rather than maximizing prediction performance.





Random Forest

Although less interpretable than a single decision tree, the Random Forest model highlights consistent patterns across many trees. The variable importance analysis identifies ***ST_Slope*** as by far the most influential predictor, followed by



ChestPainType, ***MaxHR***, and ***Oldpeak***, which aligns closely with the results obtained from Logistic and Penalized Logistic Regression. This consistency strengthens confidence in the relevance of these predictors for heart disease classification. The out-of-bag error (OOB) is **13.9%**, corresponding to an accuracy close to **0.87–0.88**, which is comparable to the best-performing logistic and Naive Bayes models, and it stabilizes as the number of trees increases, suggesting a robust and reliable model.

4. Feature Selection

Lastly, we studied how model performance changes when using different subsets of predictors instead of the full set. Using the same Random Forest variable importance computed in the previous section, we selected the six most relevant predictors (ST_Slope, ChestPainType, MaxHR, Oldpeak, Cholesterol and Age) and used them to build a reduced logistic regression model.

Using this information, we defined three logistic regression models and evaluated all of them on the same test set. The full model includes the complete set of 15 predictors (original variables plus the engineered HighRisk, HighRiskAngina and HighRiskST) and achieves accuracy 0.879, AUC 0.933 and Kappa 0.755, so it is still the model with the best predictive performance. The Top6 RF model uses only the six variables selected by the Random Forest and still reaches accuracy 0.868, AUC 0.902 and Kappa 0.735, showing that most of the predictive information is concentrated in a relatively small group of features.

Finally, the risk-focused model is built only with clinically meaningful variables and the engineered risk indicators (ChestPainType, ST_Slope, ExerciseAngina, Oldpeak, HighRiskST and HighRiskAngina) and attains accuracy 0.846, AUC 0.907 and Kappa 0.691. Although its accuracy and Kappa are slightly below the full model, it remains highly competitive and is easier to interpret from a medical point of view, since all predictors have a clear clinical meaning. Overall, these results illustrate the usual bias–variance trade-off: reducing the number of predictors simplifies the model and may improve robustness, while only slightly decreasing performance, which is exactly the kind of feature-selection behaviour discussed in the course.



	Model	Accuracy	AUC	Kappa
1	Full (15 vars)	0.879	0.933	0.755
2	Top6 RF (6 vars)	0.868	0.902	0.735
3	Risk-focused (6 vars)	0.846	0.907	0.691

Conclusions: Overall, the project shows that heart disease can be predicted with high accuracy using standard supervised learning tools and careful data preparation. After cleaning implausible values, imputing missing cholesterol and engineering simple risk indicators, we obtained models with AUC values around 0.93 and accuracies close to 0.88 on the test set, which indicates very strong discrimination between patients with and without heart disease. Logistic regression, its penalized version and Naive Bayes performed best, while simpler models such as LDA, QDA, KNN and decision trees were slightly worse but helped to confirm the relevance of key variables like ST_Slope, ChestPainType, MaxHR and Oldpeak. Finally, the feature-selection analysis showed that a reduced and clinically meaningful set of predictors can retain most of the predictive power of the full model, providing a good balance between accuracy, robustness and interpretability for potential real-world use.